

Education

University of Utah
B.S. Computer Science

Salt Lake City, UT
August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies
Associate Software Engineer

Dallas, TX
June 2025 – Present

- Reduced manual validation overhead for defense sensor hardware by building automated CI/CD pipelines in Python and C++ with static analysis and integration testing, enabling engineers to ship to embedded targets with confidence across multiple production configurations.
- Improved data pipeline reliability for high-frequency sensor ingestion by designing modular, fault-tolerant microservice architectures on AWS, making engineering data programmable and queryable across hardware variants rather than buried in one-off scripts.
- Embedded directly with hardware and firmware engineers as their primary software counterpart, translating messy hardware constraints and vendor interfaces into clean service abstractions the broader team could depend on.

Doxy.me
Software Engineer

Charleston, SC
August 2024 – May 2025

- Replaced manual clinical data workflows by building LLM-powered Python and TypeScript pipelines that automatically extracted and structured data from unstructured inputs, reducing review overhead and enabling downstream analytics at scale.
- Integrated multiple complex vendor APIs including WebRTC for real-time video streaming and AWS SageMaker for managed inference, writing production-grade adapters that normalized messy vendor interfaces into a consistent internal data layer.
- Improved data versioning and deployment reproducibility across production environments by owning YAML-based CI/CD configuration and infrastructure automation, ensuring every release was validated, traceable, and rollback-safe.

Projects

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.